

Natural Intelligence: A Biomimicry Art and Science Installation

Maya Angia

Abstract. Status: concept. Nothing has been built. Natural Intelligence is a proposed multi-element installation arguing a single thesis: nature has already solved the problem. It would bring biomimicry, the practice of learning from and emulating biological systems to solve human design challenges, off the page and into a lived campus experience through three interlocking elements: a mural whose technical-drawing half dissolves into biological illustration, a tactile decision tree that dispenses interlocking 3D-printed pieces so each visitor assembles and keeps a personal sculpture representing the path they reasoned along, and a problem wall where visitors deposit real problems for others to pull and run through the tree. The installation is designed to be complete in thirty seconds or thirty minutes, and legible to engineers, artists and everyone in between. This document sets out the concept, siting, evaluation plan and timeline as proposed.

Keywords. biomimicry; public installation; science communication; participatory design; digital fabrication; bio-inspired design

1 Thesis

Biomimicry is usually presented either as a list of finished products, Velcro and the bullet train and the kingfisher beak, or as a design philosophy stated abstractly. Neither conveys the part that actually transfers, which is the reasoning procedure: take a problem, abstract it to a function, find organisms that solve that function, and translate the mechanism rather than the form.

The installation's proposition is that this procedure can be made physical, and that a visitor who walks it once will recognise it afterwards. Every element makes the same argument by a different route.

2 Elements

2.1 Mural

A large-scale painting, temporary siting acceptable, with one half rendered in scientific and technical drawing language (architectural blueprints, Fibonacci spirals, fractals, reaction–diffusion systems) merging into organic biological illustration. A Voronoi diagram becomes a giraffe's coat. A tensegrity wireframe becomes a cell cytoskeleton. A branching optimisation network becomes a lung. At the centre the two visual languages bleed together, and the solution and the organism become the same thing, which is the thesis made visual rather than stated.

Integrated border panels ground the art in current work: Velcro from burdock, the Shinkansen nose from the kingfisher beak, Opteran's insect-brain neural networks, the slime mould solution to the Tokyo rail network.

2.2 Decision tree

A tactile, illustrated pathway walking visitors through biomimicry logic. It opens with one question, what kind of problem is being solved, and branches by constraint type: strength with minimal material, moving

through fluid, self-repair, efficient distribution. Each branch runs three to four steps, each step illustrated with a different organism.

The mechanic that makes it work: at every node along a path, the visitor collects a small 3D-printed piece corresponding to the structural principle at that node. A Voronoi cell. A tensegrity rod-and-cable unit. A logarithmic spiral segment. A fractal branching fork. A honeycomb tile. All pieces share one push-fit connector geometry, so they assemble regardless of which path produced them. By the terminal node a visitor holds three to five pieces that combine into a personal sculpture they keep, with each piece labelled on the reverse with its organism and its principle. The takeaway is a physical record of how a problem was reasoned through, not a souvenir.

The tree is traversable in reverse: pick an organism's trait and find out what problem it could solve. No two paths yield the same sculpture, and the same path run backwards can yield a different problem.

2.3 Problem wall

Visitors write down real problems they want solved. Others pull a card and run it through the decision tree, producing crowdsourced, biomimicry-derived answers to strangers' problems. The wall fills and evolves across the run, becoming its own living artwork and data visualisation. At the close, the full archive of problems and the solutions they generated would be displayed as a final piece in itself.

3 Siting and public sharing

The installation would occupy a high-traffic campus location chosen specifically for cross-disciplinary foot traffic: BioQuad, the Student Center, Tech Green and Skiles Walkway, or the library. It should stand for several weeks during the fall semester to maximise the number of students, faculty and visitors who encounter it without seeking it out.

Sharing would run on several channels beyond the installation itself: Georgia Tech student social media, the CBID newsletter, the fall bio-inspired design course, and the Colleges of Engineering and of Design. Outreach to student organisations with design and videography presence would extend reach further. A companion page would document the problem wall's evolution and the diversity of take-home sculptures, creating an ongoing digital extension of the physical piece.

4 Evaluation

Success has three layers, measured differently.

1. **Reach.** Count unique interactions via take-home sculptures distributed, each of which is individually countable, and problem cards deposited and pulled. A target of 100 or more meaningful interactions is realistic for several weeks in a high-traffic placement.
2. **Breadth.** Photograph and archive the problem wall weekly. The disciplinary spread of problems deposited is the measure of whether the installation reached beyond the people who already care about biomimicry, which is the only audience worth measuring. A QR-code survey would capture what visitors took away and whether they learned something unexpected.
3. **Propagation.** Each sculpture leaves with a person and carries the thesis with it. A small QR code on the base piece would link to a page where people can record what problem they were solving and what they built, making the propagation measurable rather than assumed.

A project journal would capture design decisions, fabrication problems and community response, as source material for later biomimicry work.

5 Collaboration

Dr. Jeannette Yen (Center for Biologically Inspired Design, Georgia Tech) is being approached as a content consultant, principally for biological accuracy review of the mural and decision tree. **This has not yet been discussed with her.** Conversations with biomimicry-forward companies and Atlanta design firms as sponsors, speakers or volunteers are at an early stage and would supplement rather than depend on external funding.

6 Timeline as proposed

Window	Activity
Early August	Finalise concept, confirm installation space, begin outreach to consultants and sponsors
Mid August	Finalise mural design and decision tree content, complete biological accuracy review, order materials
Late Aug – early Sep	3D print all modular sculpture components, print educational border panels
Mid September	Begin mural painting; assemble problem wall components
Late September	Mural complete; soft launch for peer and consultant feedback
Early October	Incorporate feedback, finalise all elements
Mid–late October	Installation goes live, three to four weeks of active display
By 1 December	All funds expended, documentation complete

7 Requested support

- Help securing a high-traffic covered or indoor wall that can remain in place through late November.
- Access to campus fabrication facilities (Invention Studio, HIVE, or College of Design shops) for the modular sculpture components.
- Introductions to Atlanta-area muralists or scientific illustrators working at the intersection of art and science, which would raise the mural’s visual quality and could become a genuine collaboration.
- Connections to biomimicry-forward companies with Atlanta presence who might sponsor materials, contribute to an opening event, or co-brand the installation.

8 Relevant preparation

Preparation for this project spans both sides of what it asks. On the science side: two years of research in biomimicry-adjacent fields including bio-robotics, neuromorphic systems and medical device design, plus formal training in bio-inspired design, which supplied a rigorous framework for translating biological strategies into design principles. On the art side: prior experience painting a public mural, so composition planning at scale, managing paint in a public space and carrying a piece from sketch to finished wall are not new. Digital fabrication experience with 3D printing comes from coursework, research and independent projects.

Two things would be genuinely new. Designing and executing a fabricated sculpture system and a designed physical experience is unfamiliar territory, and communicating a scientific framework through visual and tactile language rather than through data or writing is a different skill from the one this portfolio

otherwise demonstrates. Initiating and managing a project with this much community interaction is also new, and the problem wall in particular requires thinking like a community organiser rather than an engineer.

9 The concept set this was selected from

The proposal above is one of seventeen concepts developed during ideation. Several of the others have independent merit and are recorded here in full, both because the selection reasoning is part of the design work and because some of them are better suited to a different brief than this one.

9.1 Selection criteria

Six criteria were applied, and the final concept is the only one that satisfied all six.

1. **It must argue the thesis, not merely look biological.** Beautiful biological imagery is not the same as demonstrating that biology solves problems. This criterion eliminated the largest number of otherwise strong concepts.
2. **Completable in thirty seconds or thirty minutes.** A piece that only rewards sustained attention loses the cross-disciplinary audience that justifies a high-traffic siting.
3. **Something leaves with the visitor.** Physical propagation is the only honest measure of whether an idea transferred.
4. **No screen, no app, no download.** Any concept requiring a device between the visitor and the work adds a barrier and dates immediately.
5. **Fabricable with campus facilities on an undergraduate budget.**
6. **It accumulates.** A static piece is the same on day one and day twenty-one; a piece that grows rewards return visits and produces a record.

9.2 Concepts using a hidden or revealed layer

Five concepts shared a single mechanic: an ordinary surface concealing a second layer the visitor uncovers.

UV ink mural, “The Secret Biology.” Clean minimal line art in daylight, with an entire hidden layer in UV-reactive ink (veins, nerve pathways, cellular structures) appearing only under a blacklight wand mounted on a cord beside it. Simple, striking, no app required.

Lenticular science panels. Panels showing one abstract image straight on that shift to a microscopy image, anatomical diagram or biological pattern when viewed at an angle. The effect is pure physics. Mounted in sequence along a hallway, walking past tells a visual story.

Temperature-reactive mural. Thermochromic paint that looks like one thing at room temperature and reveals a hidden pattern, phrase or cellular structure where a hand is pressed or breath lands. Deeply tactile, and the interaction *is* the reveal.

Shadow anatomy. A relief wall piece designed so that at one specific time of day its cast shadow forms a recognisable biological shape: a heart, a neuron, a DNA helix. The piece itself reads as abstract. A plaque saying only “come back at 2pm” does the rest.

Embedded cipher. A typographic installation where the visible text reads as a poem, with a steganographic layer, readable via a printed decoder card, carrying the actual science underneath: equations, data, a research finding. Two audiences, one piece.

Why set aside. All five are genuinely good, and all five fail criterion one. A hidden layer creates delight and rewards curiosity, but the thing revealed is an image rather than an argument. The visitor leaves having

discovered something, not having reasoned through anything, and nothing leaves with them. The shadow anatomy concept fails criterion two badly, since it works for roughly twenty minutes per day. These are strong candidates for a gallery brief rather than a thesis-driven one.

9.3 Bio art concepts

Reaction–diffusion wall. Reaction–diffusion systems generate leopard spots and zebrafish stripes, and almost nobody knows the patterns come from biology. A large mural of these patterns with panels showing the corresponding species and the Turing equations that generate them.

Living art, a petri dish gallery. Pigment-producing bacterial or fungal strains cultured on agar and guided into shapes or gradients, then sealed, preserved and framed. Each piece is grown rather than painted, with a short making-of video looping nearby. This is bioart in the most literal sense.

“Morphogenesis” textile installation. Large hanging fabric panels hand-dyed using techniques that mimic biological self-organisation: diffusion dyeing, resist patterns echoing branching vasculature, tie-dye governed by mathematical rules rather than randomness. Hung in an atrium so they move with airflow.

Why set aside. The reaction–diffusion concept survived longest of any alternative and its central move, pairing a pattern with the equation that produces it and the organism that wears it, was absorbed directly into the final mural. Petri dish culture introduces biosafety approval and a live-specimen timeline that an undergraduate budget cannot absorb, and preserved plates degrade unpredictably. The textile concept is beautiful and fails criteria one and three completely.

9.4 Interactive sculpture concepts

Magnetic tensegrity sculpture garden. Tabletop or standing tensegrity structures, held together purely by tension so the rods appear to float, with rare-earth magnets at the connection points. Visitors reconfigure the tension paths, collapse and rebuild them, or add and remove elements. Tensegrity governs both cell mechanics and architecture, which a nearby panel explains. Nobody walks past this without touching it.

Modular Voronoi wall. Interlocking Voronoi-tessellated tiles, the geometry behind giraffe spots, dragonfly wings and bone microstructure, which visitors can remove, rotate or swap to create new configurations. Each tile carries an icon on the reverse naming the natural structure it corresponds to. A puzzle with no wrong answer.

Sand Chladni table. A steel or acrylic plate on a frame with a speaker beneath it. Visitors dial through frequencies and sand on the plate self-organises into standing-wave patterns in real time, a completely different figure at every frequency, with a panel connecting the same patterns to cymatics, embryonic development and acoustic engineering.

Why set aside. These are the three strongest alternatives and two of them contributed directly. The reverse-side labelling from the Voronoi wall became the labelling scheme on the take-home pieces, and the reconfigurable magnetic joint from the tensegrity garden became the push-fit connector geometry. Both fail criterion three on their own, since the pieces have to stay for the next visitor. The Chladni table is the one genuine loss: it is mesmerising, it accumulates nothing, it needs mains power and constant supervision, and sand migrates. It deserves to be built as a standalone piece.

9.5 Concepts using the campus as input

“GT in Motion,” a live data sculpture. A physical display, an LED matrix or a mechanical flip-panel board, in a high-traffic spot visualising real-time anonymised campus data as an abstract shifting pattern: foot traffic density, library occupancy, dining hall load. The data is mundane and the visualisation is not. A small legend allows decoding.

“How Are You, Really?” an emotional mosaic. A large grid where students place a coloured tile or dot from a dispenser according to a single question and a simple colour-mood key. The mosaic evolves over

weeks, and the full dataset is photographed and displayed as a final piece alongside visualisations of how campus mood shifted.

Campus soundscape visualisation. Ambient audio recorded from eight to ten locations, a busy intersection, the library, a rooftop, a lab, a practice room, run through audio-visualisation software to generate a unique abstract print from each recording. Displayed together, each labelled only with a location and a time.

Why set aside. All three satisfy criterion six better than the final concept does, and the accumulation logic of the emotional mosaic is what became the problem wall, which is the element most likely to surprise. But none of them is about biology at all. They are participatory data pieces that happen to be beautiful, and a biomimicry brief is the wrong home for them. The soundscape concept additionally requires recording in shared spaces, which raises consent questions disproportionate to its payoff.

9.6 The two concepts that became the proposal

“Morphology” decision tree with buildable sculpture. A large illustrated branching decision tree on freestanding panels, opening with one question about what kind of problem is being solved, branching by constraint type, each branch running three to four illustrated steps. At each node the visitor collects a 3D-printed piece corresponding to that node’s structural principle, all sharing one push-fit connector geometry, assembling into a personal sculpture by the terminal node.

“Blueprint / Organism” mural with sculpture. A mural in a hybrid visual language, half technical and architectural linework and half organic biological illustration, the two bleeding together at the centre, paired with a modular sculpture on a low plinth whose hand-fabricated pieces represent the principles depicted above and can be freely reconfigured, changing shape across the day depending on who handled it last.

These were the only two concepts satisfying criteria one through five, and each covered the other’s weakness. The decision tree carries the reasoning procedure but is visually flat; the mural is arresting but passive. Combined in one space, with the mural on the back wall, the reconfigurable sculpture centred on a plinth, the decision-tree kiosk to one side, and the illustrated field guide panels integrated into the mural border, every element reinforces the same thesis. Adding the problem wall from the campus-data family supplied criterion six, and turned the piece from a demonstration into something that answers strangers’ actual problems.

Document note. Nothing described here has been fabricated or installed. All dates, collaborations, budgets and outcomes are proposed. Section 9 records the full concept set this proposal was selected from, several of which remain viable as independent projects.